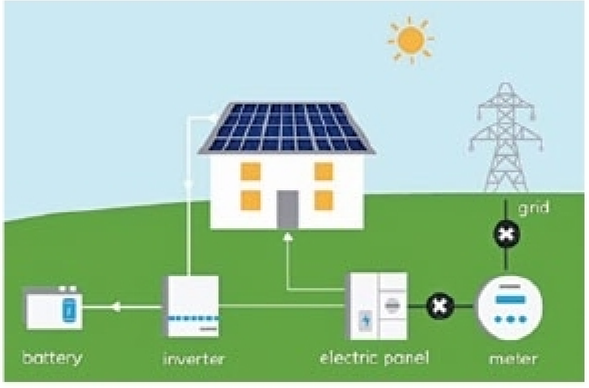

A depiction of how solar rooftops work
(Credit: energysage.com)

A grid-connected system is wired to the nearest electricity grid of the utility. The most economical grid-connected setup is done via net-metering. In case of net-metering, the user can choose to give the excess solar energy generated to the utility through the grid and, in turn, get compensated, either monetarily or by reusing the excess energy later. However, net-metering comes with legal compliances and compatibility with the utility company or discom. The user may choose to add a storage battery with this setup.

A solar rooftop system consists of many components, including solar panels that capture solar energy, mounting racks to hold the panels, inverter to convert direct current generated through the panels to alternating current, utility meter to measure electricity generation, battery pack, charge controller and some other components. Cabling makes the connections complete.

Power generation of a solar setup is measured in kilowatts (kW). To install 1kW solar setup, about 12sqm (130sqft) area is required.

Benefits of Solar Rooftops

The biggest benefit of a solar installation is the reduced dependency on the grid. At an individual level, this has long-term returns on expenses generated through energy. While electricity bills can be brought down drastically, schemes like net-metering can provide the opportunity for higher money returns through excess energy generation.

Maxson Lewis, managing director, Magenta Power

At a social level, the biggest benefit of using renewable energy is taking the load off exhaustible resources. This allows utility companies to distribute electricity more consistently